CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

December 2012

MA101 Engineering Mathematics-I

Date: 03.12.2012, Monday | Time: 01:30 p.m. To 04:30 p.m. | Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q 1(a) Give the conditions for non-homogeneous system $AX = B$ with n variables **[02]**

(1)To get unique solution

(2)To get infinite number of solution

(b) True or false: If positive term series $\sum u_n$ is convergent then $\lim_{n\to\infty} u_n = 0$. The converse of the above statement is true or not? Justify your answer. **[05]**

Q 2(a) Find the inverse of the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & -1 & 2 \end{bmatrix}$ using gauss-Jordan method if, it exists. **[05]**

(b) Show that the equations $x + 3z = kx, -5x + 4y = ky, -3x + 2y - 5z = kz$ have non – trivial solution then $k^3 - 12k + 14 = 0$. **[05]**

(c) Disscuss the convergence of the series **[04]**

(i) $5 - 4 - 1 + 5 - 4 - 1 + \cdots$

(ii) $\sum_{n=1}^{\infty} \sqrt{\frac{n}{(n+1)}}$

OR

Q 2(a) Find interval of convergence of the series $\sum_{n=1}^{\infty} \frac{n^2+1}{n^2} x^{n-1}$ **[04]**

(b) Test the convergence of the following series **[05]**

(i) $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \cdots$

(ii) $\sum_{n=1}^{\infty} \frac{e^{-tan^{-1}n}}{n^2+1}$

(c) Find rank and nullity of the matrix $\begin{bmatrix} -1 & 0 & 1 \\ 1 & 2 & 3 \\ 1 & 0 & -1 \end{bmatrix}$ **[05]**

Q3(a) State De Moiver's theorem and Simplify $\frac{(cos2\theta+isin2\theta)^{\frac{2}{3}}(cos\theta-isin\theta)^2}{(cos3\theta-isin3\theta)^2(cos5\theta-isin5\theta)^{\frac{1}{3}}}$ [04]

(b) Find the roots common to the equations $z^4+1=0$ and $z^6-i=0$. [05]

(c) Test the consistency of the system [05]

$$x+y-z=0, 2x-y+z=3, 4x+2y-2z=2$$

and if it consistent find the solution by Gauss Elimination method

OR

Q 3(a) Discuss the convergence of the series $\sum_{n=1}^{\infty}\frac{n^p}{(n+1)^q}$ for different values of p and q. [04]

(b) Prove that the n^{th} root of unity are in geometric progression and also show that their sum is zero. [05]

(c) If $x_n = cos\frac{\pi}{3^n} + i\,sin\frac{\pi}{3^n}$, show that [05]

(i) $i(x_1 \cdot x_2 \cdot x_3 \cdots x_\infty) = -1$ and (ii) $i(x_0 \cdot x_1 \cdot x_2 \cdots x_\infty) = 1$

SECTION-II

Q4(a) If $y=(2-3x)^{10}$, then find y_9. [01]

(b) Expand $f(x)=3x^3+9x^2+5$ in power of $(x-1)$. [03]

(c) If $u=\log(x^2+y^2)+\tan^{-1}\left(\frac{y}{x}\right)$, then show that $u_{xx}+u_{yy}=0$ [03]

Q5(a) If $u=\tan^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, then find [04]

(i) $x^2u_{xx}+2xyu_{xy}+y^2u_{yy}$ and (ii) xu_x+yu_y

(b) Evaluate $\lim_{x\to 0}\frac{\log x}{\cot x}$. [02]

(c) If $z(x+y)=x^2+y^2$, then show that $(z_x-z_y)^2=4(1-z_x-z_y)$. [04]

(d) Find the maximum value of xyz subject to $\frac{x^2}{25}+\frac{y^2}{16}+\frac{z^2}{9}=1$ [04]

OR

Q5(a) If $u=\cot^{-1}\left(\dfrac{x^3+y^3}{x^2+y^2}\right)$, then find $x^2u_{xx}+2xyu_{xy}+y^2u_{yy}$ and xu_x+yu_y. **[04]**

(b) Evaluate $\lim_{x\to\frac{\pi}{2}} \dfrac{\tan x}{\tan 3x}$. **[02]**

(c) If $f(x, y)=\dfrac{1}{xy}+\dfrac{1}{x^2}+\dfrac{\log x-\log y}{x^2+y^2}$, then show that $xf_x+yf_y+2f=0$. **[04]**

(d) Find the maximum value of $x^2y^3z^4$ subject to $2x+3y+4z=9$ **[04]**

Q6(a) Test the relative maxima and minima for $f(x,y)=x^3+y^3-3x-12y+20$. **[04]**

(b) If $u = x^2 - y^2$ where $x = e^t cost$ & $y = e^t sint$ then find value of $\frac{\partial u}{\partial t}$. **[02]**

(c) Find the approximate value of $\sqrt{(299)^2+(399)^2}$. **[04]**

(d) Find the equation of tangent plane and normal line of $x^2+2y^2+3z^2=12$ at (1, 2, -1) **[04]**

OR

Q6(a) Test the relative maxima and minima for $f(x,y)=x^3+3x^2-4xy+y^2$. **[04]**

(b) If $u=x^2+2xy+y^2$, where $x=t\cos t$ & $y=t\sin t$, then find value of $\dfrac{du}{dt}$. **[02]**

(c) Find the approximate value of $\sin 44^\circ \cos 62^\circ$. **[04]**

(d) Find the equation of tangent plane and normal line of $z=e^{-x}\sin y$ at $\left(0, \dfrac{\pi}{2}\right)$. **[04]**

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